Predictions

1

a Interpolation b Extrapolation c Interpolation d Extrapolation

2

a $y = -23.6079$ b $y = 67.8912$ c $y = 36.056$ d $y = 8.7592$

3 50.536 cm

4

a $y = 0.5x + 5$ b 8.45

5

a $y = -1.2x + 16$ b $y = 8$

6

a $y = 13.44$

b The data is strongly correlated and the prediction is an interpolation, so the prediction is reliable.

7

a $y = 32.16$

b The data is weakly correlated and the prediction is an extrapolation, so the prediction is unreliable.

8

a $y = 38.53$

b The data is moderately correlated and the prediction is an extrapolation far from the data range. There's no telling that the linear trend continues so the prediction is unreliable.

9

a $y = 94.04$

b The data is moderately correlated and the prediction is an interpolation, so the prediction is moderately reliable.

10

a $y = -45.69$

b The data is strongly correlated and the prediction is an extrapolation close to the data range, so the prediction is reliable.

Applications

11

a Yes, because it is interpolation.

b 2448 Pa

c

i Yes ii No iii No iv No

12

a 90

b 22°C

c

i No ii Yes iii No iv No

13

a 3 seconds

b 2.5 hours

14

a 20 years

b 1300 fish

c 11 years

15

a 8500 m

b 6500 m

c 300 seconds

d 1800 seconds

16

a 20°C

b 210 chirps per minute

c 130 chirps per minute

17

a 48 m

b $y = 3x + 30$

c 52.5 m

d Yes, because it is using interpolation.

18

a $y = -0.005x + 49$

b $y = 29^{\circ}C$

c Yes, because it is using interpolation.

19

a Negative gradient

b -2.5

c $y = -2.5x + 105$

d $y = 95$

20

a $\frac{1}{16}$

b $c = 100$

c $\$400\,000$

21

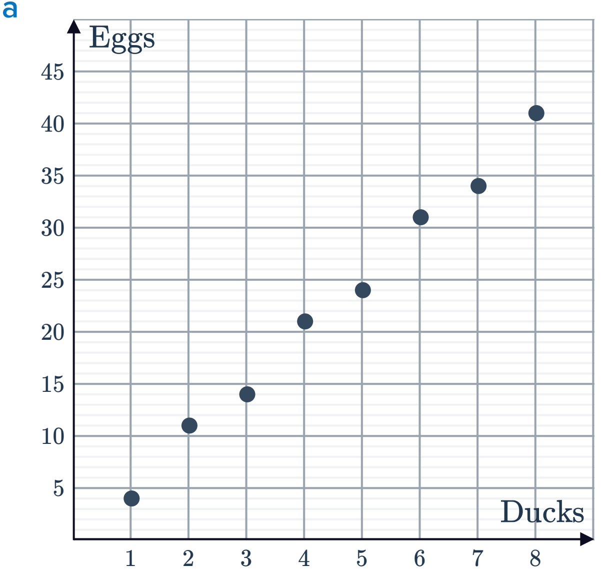
a $A = -25T + 750$

b $A = 100$ hectares

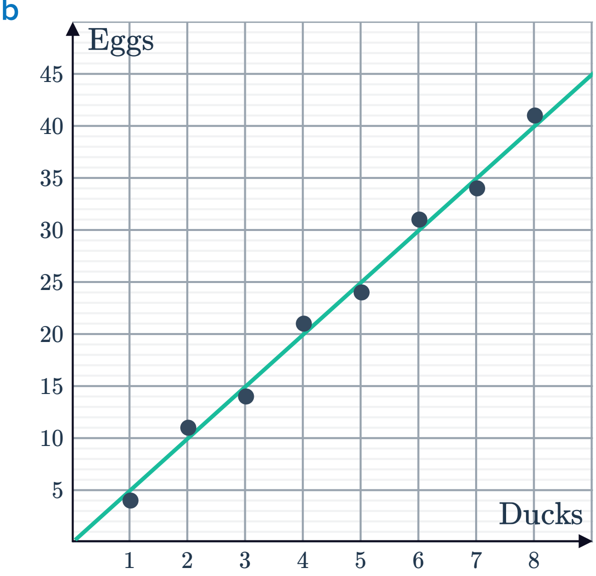
c No, because it is using extrapolation.

22

a



b



c 5

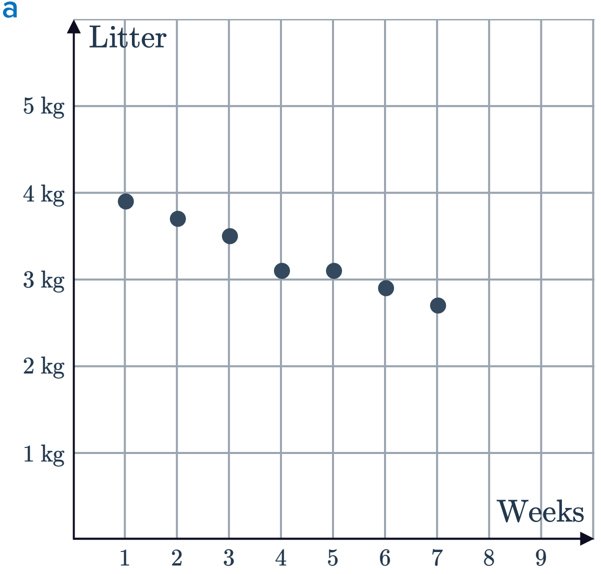
d 0

e $y = 5x$

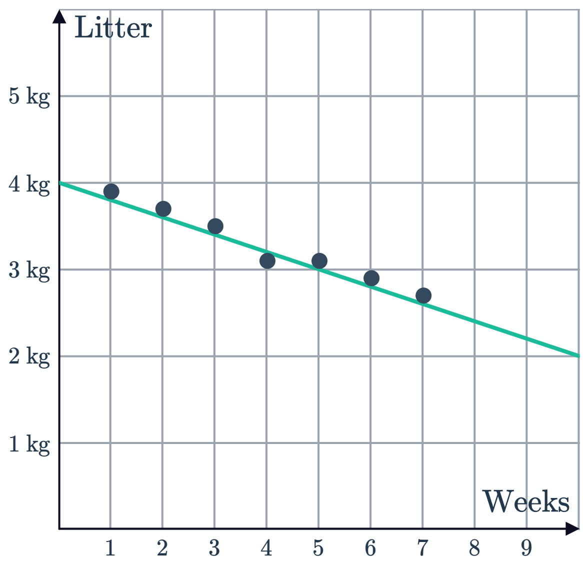
f 150 eggs

23

a



b Example answer:



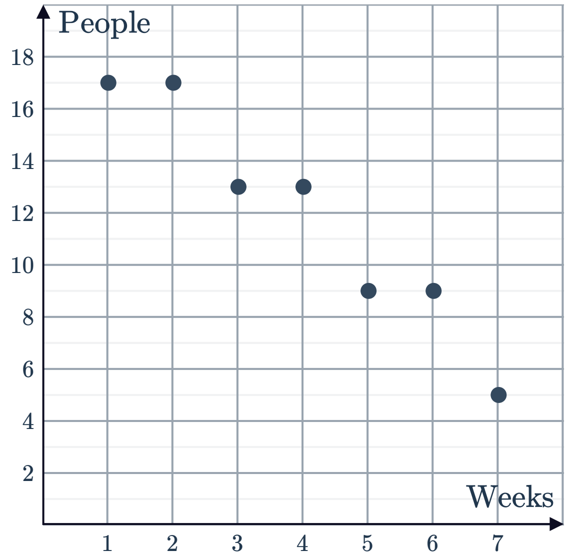
c Example answer: $y = -0.2x + 4$

d Approximately 2.4 kg

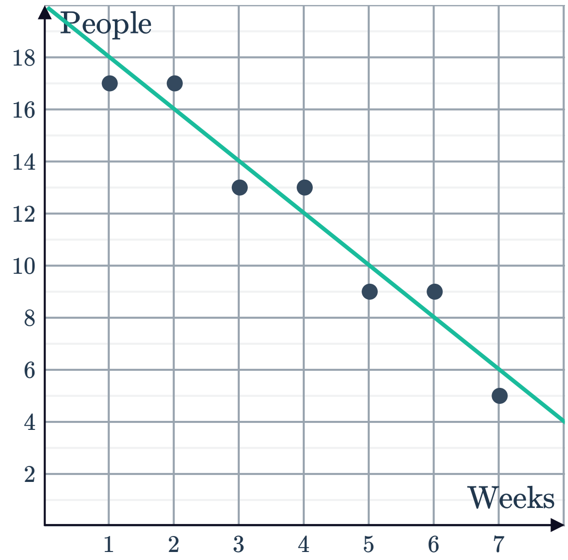
e No, because it is using extrapolation.

24

a



b Example answer:



c Example answer: $y = -2x + 20$

d Approximately 0

25

a $14\,811$

b The y-intercept indicates that when a house is brand new, its value is, on average, $\$14\,811$.

c No. When a house is zero years old its age is not within the data range.

d Approximately 4.9 years.

e $\$1\,200\,000$

26

a The y-intercept indicates that when a student watched no TV, their average mark in the exam, was 90.

b Yes. When a student watches 0 hours, their time is within the data range, and the mark makes sense.

c No

27

a $18\,000$

b The y-intercept indicates that when a Mitsubishi Lancer is brand new, its value is, on average, $\$18\,000$.

c Yes. When a car is zero years old its age is within the data range.

d 11 years

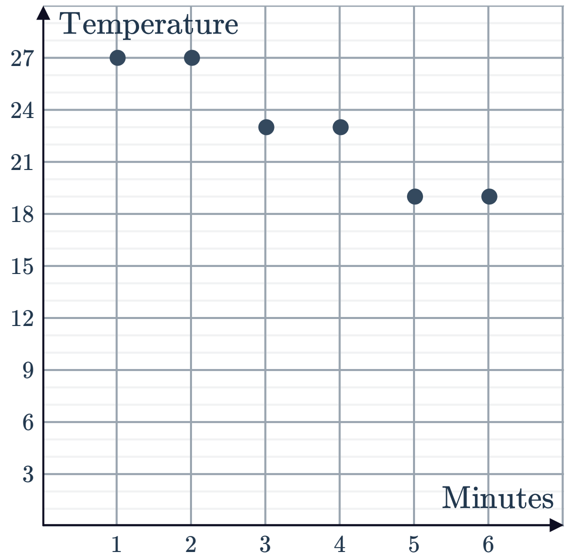
e No. When a car is older than 11 years, the line goes below the x-axis, which would give a negative value, which is not possible. We cannot use the line to find the value since this age is not within the data range.

a $y=-3x+60$

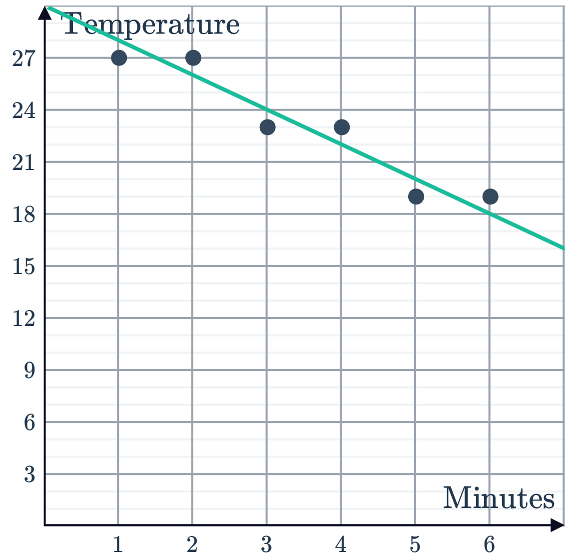
b 24

29

a



b



c -2

d 30

e $y=-2x+30$

f $x=\dfrac{15}{2}$

30

a For each 1000 km increase in distance driven, the least-square line predicts that the maintenance costs increase by $\$990$.

b A car that is not driven at all is predicted to cost $\$14620$ to maintain.

c No, because for a car that hasn't been driven, it should have little to no cost to maintain.

d $\$50\,260$

e Yes, because it is an extrapolation close to the data range where the cost is likely to increase along with distance driven, and the correlation coefficient is high.

f Yes, a car that is driven more will have more wear and tear and would be likely to have greater maintenance costs.